DEVELOPMENT OF AN INTERACTIVE ASSESSMENT DEVICE

TO RECORD USER HAND MOVEMENT WITH THE DISPLAY OF MOTION & FORCE DATA DURING THE ADL REHABILITATION

(PROJECT ID: 1M21372)

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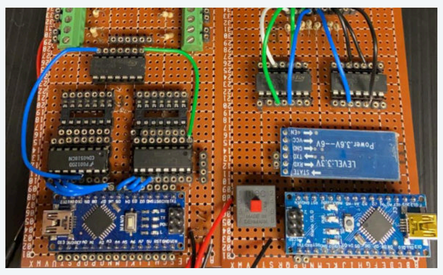
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ABSTRACT

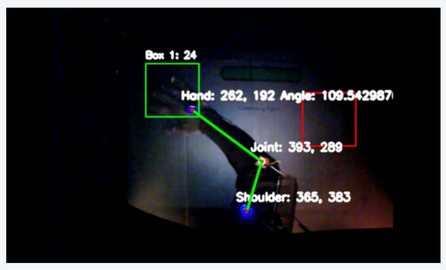
The project aims to provide an interactive approach to objectively and quantitatively measure the Functional Status for one's Activities of Daily Living abilities, which refers to fundamental activities needed for an individual to function independently in everyday life. Additionally, collating those preliminary measurements to aid in the curation of an Asian-centric Normative Movement Database.

DESIGN/METHODS



FORCE DATA ACQUISITION

- **Multiplexed** Force Sensitive Resistors (FSR) with **Silicon Padding** for equal pressure distribution
- Newton data points saved via **HC-05** Serial Communication



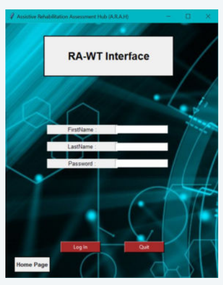
MOTION DATA ACQUISITION

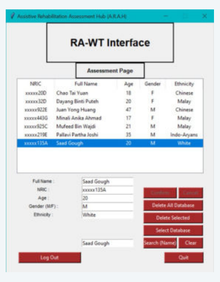
- **Optoelectronic** marker-based System
- Markers mapped with **Computer Vision**
- **Trigonometric** Concept; Motion Capture Analysis
- **.mp4** playback capabilities; 480p, 30FPS



AUGMENTED REALITY GAME-TASK

- **Pygame**; Presentation Agent
- **Computer Vision**; Logic Interface
- **Threading**, Parallel Processing





GRAPHIC USER INTERFACE (GUI)

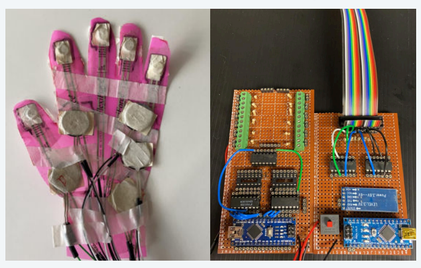
All-in-One Interface:

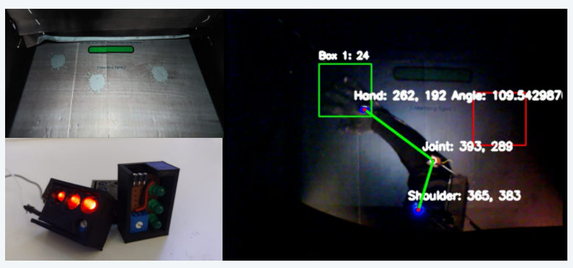
- Tkinter, Treeview, Matplotlib (**Front-End**)
- SQLite3 (**Back-End**); Patient Record
- Patient Local **File Management**
- Assessment **Playback**

DICUSSION/EVALUATION

FORCE DATA - TOTAL NEWTON APPLIED FROM HAND

- Exhibit an **inverse correlation** with **pressure applied** against **internal resistance**.
- Silicon Padding utilized for **normalized pressure** across target site.
- The resistors were processed into Netwon using the **Voltage Follower Rule**.





MOTION DATA - ELBOW JOINT ANGLE

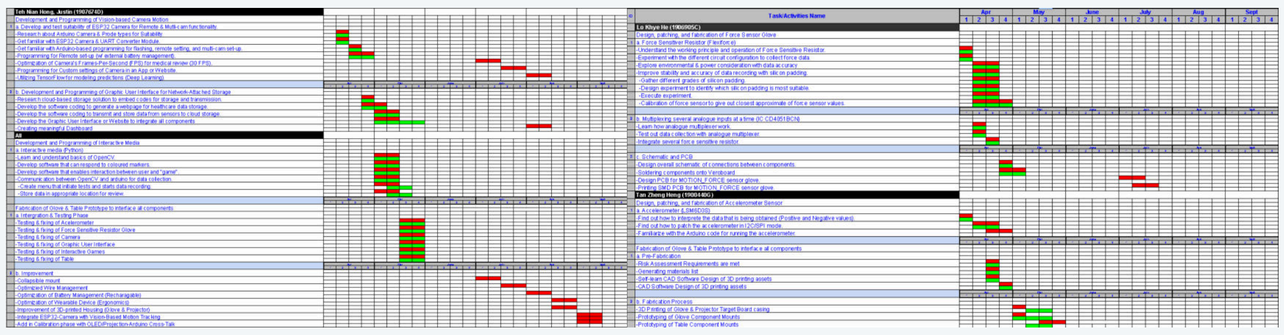
Two-Dimensional Optoelectronic Marker-based Sets typically require:

- Only **one Motion Camera**, and;
- **At least two markers** for **Sagittal plane analysis**.

The markers' underlying principle:

- Serve as **beacon points** for a Motion Camera's Computer Vision to recognize the markers (by colour) for the **motion sequence tracking**.
- **Histogram/Colour profiles** of markers are crucial for consistent tracking.

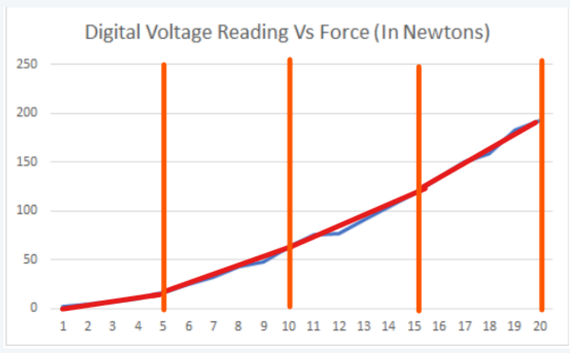
PROJECT TASK & MILESTONE

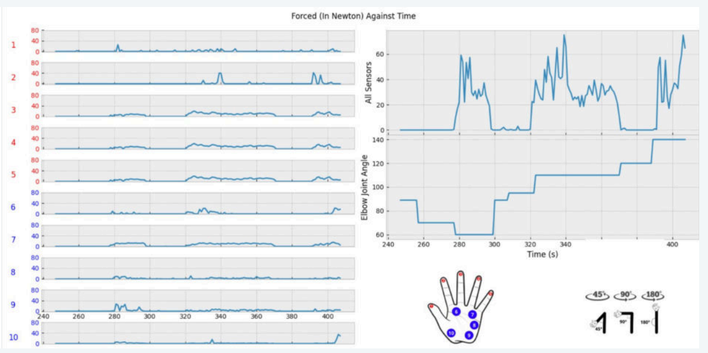


RESULTS & DATA ANALYSIS

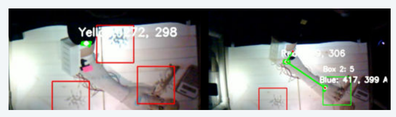
FSR Accuracy (Significant improvement; +/- 0.5 N)

- **Baseline Calibration**; Eliminate residual "voltage-noise" artefacts; Sampling & Normalizing readings.
- **Linear Segmentation**; Obtain best-fit model of Resistance-to-Newton conversion.





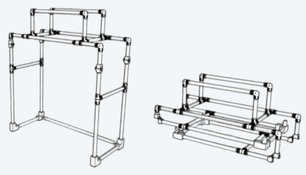
Electroluminescence (Light-Emitting Diode) Marker Sets



- Features consistent and prominent histogram profile.
- Improves marker-colour detection during motion sequence capture.
- Iterations:
  - **Stage 1**: Fine-tuning individual marker Color Threshold
  - **Stage 2**: Adopted electroluminescence-based Marker Set
  - **Stage 3**: Modification of Capture Environment
    - Curtains and vanguard cardboard retrofitted
  - **Stage 4**: Final Improvement of Marker Set Design
    - Omni-directional beaming



Additional Features



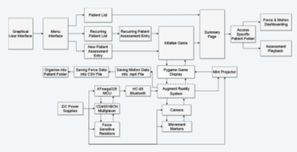
- Portability:
  - Collapsible Legs
  - Weights ~7 KG
- Interactive Augmented Reality Game-based Approach in IADL Assessment.
- Back-end Development for support of Asian-centric Normative Database curation.



CONCLUSION

The assessment device was successfully developed for its two-pronged scope of:

1. Objective Data Acquisition (Enables Qualitative Review)
   a. Combination of **Wearable** (Force Sensors), and;
   b. **Environmental-based** (Computer Vision) systems.
2. Therapy Assessment
   a. Programming of GUI App for **progressive performance data collection**;
   b. **Patented Implementation** of an **Interactive Game** with objective assessment parameters and measurements for IADL Assessment.



We envision the assessment device complementing existing therapy processes for recovery patients by objectively capturing the relevant data points over the course of the user's therapy protocol, thus quantitatively mapping out a projection of the patients' performance across different visits.

S/No	Application No.	Title of Invention	Status
1	10202105829Q	Interactive Instrumental Activities of Daily Life (IADL) Data Collection Approach with Integrated Sensors and Network-Attached Storage.	Application Received

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[1] P. Edemekong, et al. “Activities of Daily Living.” NCBI.gov. https://www.ncbi.nlm.nih.gov/books/NBK470404/ (accessed Mar. 15, 2021)

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